4-1

Subspace

- A vector set V is called a subspace if it has the following three properties:
- 1. The zero vector **0** belongs to V
- 2. If **u** and **w** belong to V, then **u+w** belongs to V

Closed under (vector) addition

- 3. If **u** belongs to V, and c is a scalar, then c**u** belongs to V

Closed under scalar multiplication

2+3 is linear combination

97 Let $T: R^n \rightarrow R^m$ be a linear transformation. Prove that if V is a subspace of R^n, then $\{T(\boldsymbol{u}) \in R^m : \boldsymbol{u}\ is\ in\ V\}$ is a subspace of R^m.

① $0 \in W$

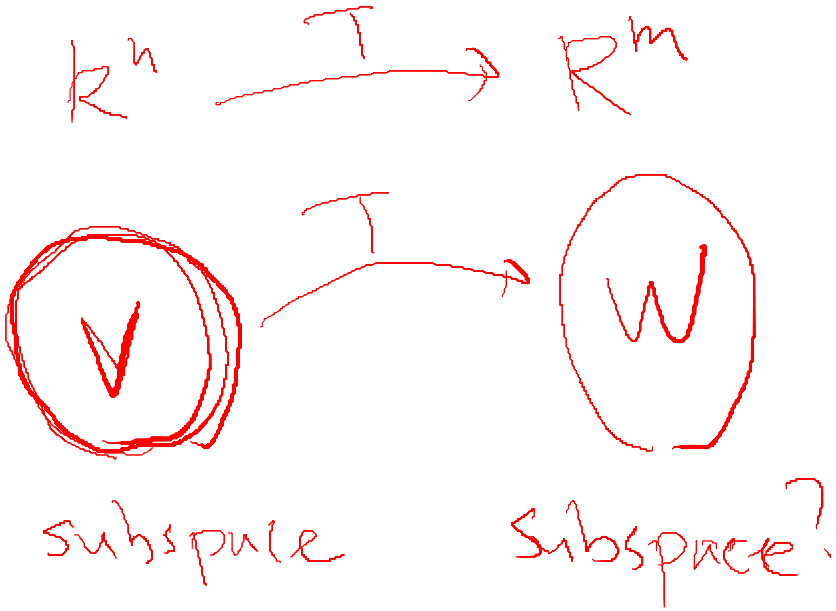
② $T(u_1) \in W,\ T(u_2) \in W$

$T(u_1) + T(u_2) = T(u_1 + u_2)$

$\because u_1 + u_2 \in V \Rightarrow T(u_1 + u_2) \in W$

③ $T(u) \in W,\quad cT(u) = T(cu) \in W$

$u \in V,\ cu \in V$



98 Let $T: R^n \rightarrow R^m$ be a linear transformation. Prove that if W is a subspace of R^m, then $\{\boldsymbol{u}: T(\boldsymbol{u})\ is\ in\ W\}$ is a subspace of R^n.

$R^n \xrightarrow{T} R^m$

① $0 \in V$

② $v_1 \in V,\ v_2 \in V$

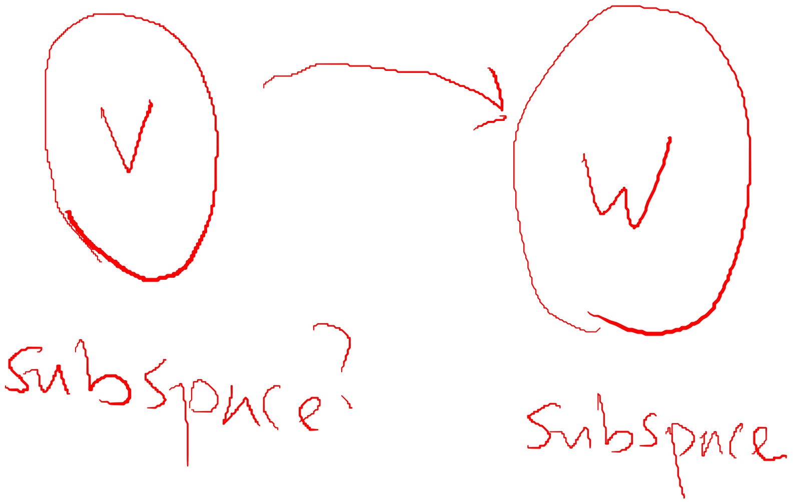
$T(v_1) \in W,\ T(v_2) \in W$

$T(v_1) + T(v_2) \in W$

$\Rightarrow T(v_1 + v_2) \in W \Rightarrow v_1 + v_2 \in V$

③ $v \in V,\ T(v) \in W,\ T(cv) \in W \Rightarrow cv \in V$

$cT(v) \in W$



99 Let A and B be two $m \times n$ matrices. Use the definition of a subspace to prove that $V = \{v \in R^n : A\boldsymbol{v} = B\boldsymbol{v}\}$ is a subspace of R^n.

① $0 \in V$? $A0 = B0 = 0$

② $v_1 \in V$, $v_2 \in V$, $v_1 + v_2 \in V$?

$Av_1 = Bv_1$, $Av_2 = Bv_2 \Rightarrow A(v_1 + v_2) = B(v_1 + v_2)$

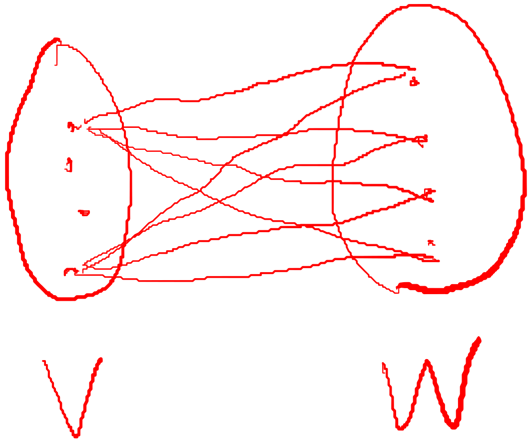
③ $v \in V$, $cv \in V$?

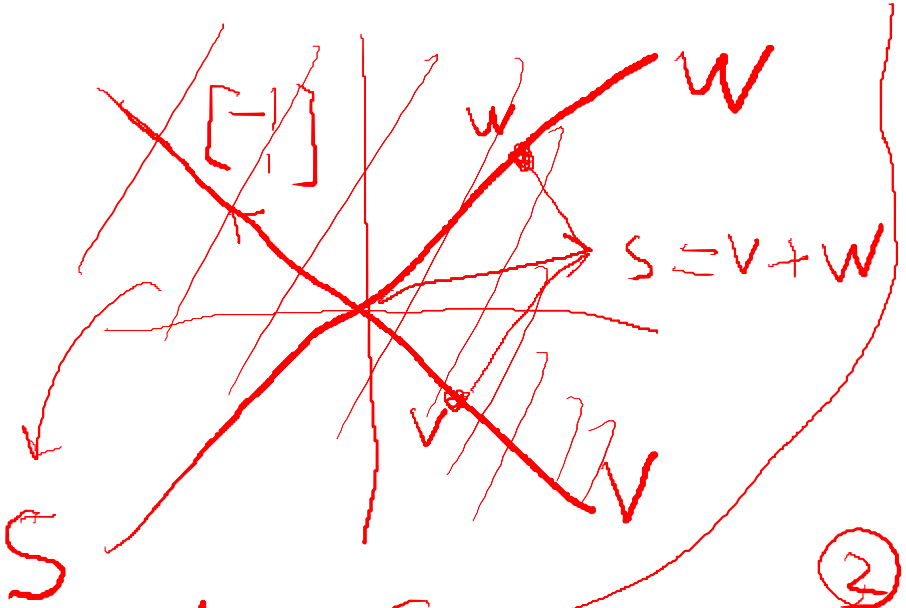
$Av = Bv \Rightarrow A(cv) = B(cv)$

100 Let V and W be two subspaces of R^n. Use the definition of a subspace to prove that

$$S = \{\boldsymbol{s} \in R^n : \boldsymbol{s} = \boldsymbol{v} + \boldsymbol{w} \ for\ some\ \boldsymbol{v}\ in\ V\ and\ w\ in\ W\}$$

is a subspace of R^n.





① $0 \in S$

$0 \in V$, $0 \in W$

$S = V + W$

$= 0 + 0 = 0$

② $s_1 \in S$, $s_2 \in S$

$s_1 + s_2 \in S$?

③ $s \in S$, $cs \in S$?

$\in V \quad \in W$

$s_1 = v_1 + w_1$

$s_2 = v_2 + w_2$

$s_1 + s_2 = (v_1 + v_2) + (w_1 + w_2)$

$\in S$

$$\begin{pmatrix}1 & & & & & 0\\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ 0 & & & & & 2\end{pmatrix} A = \begin{bmatrix} A \\ \boxed{} \end{bmatrix} \times 2$$

$$A \begin{bmatrix}1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 2\end{bmatrix} = \begin{bmatrix} A & \boxed{} \end{bmatrix}$$

$\times 2$